CHAPTER 7

FIRE AND SMOKE PROTECTION FEATURES

General Comments

This chapter provides the requirements for the inspection and maintenance of existing passive fire protection features in existing buildings and structures. In general, these requirements are intended to maintain the required fire-resistance ratings and limit the spread of fire and smoke.

Chapter 7 is divided into eight sections. Section 701 lists the general administrative provisions that require fire-resistance-rated construction, smoke barriers and smoke partitions to be maintained. Section 702 provides a list of the definitions that are used in the chapter. The remainder of the chapter describes the required specific maintenance of Penetrations (703), Joints and Voids (704), Door and Window Openings (705), Ducts and Air Transfer Openings (706), Concealed Spaces (707) and Spray and Intumescent Fire-Resistant Materials (708). These sections use the same terminology that is used in the *International Building Code*® (IBC®). As stated in Section 701.1, the requirements for the installation of passive fire protection features in new buildings are located in IBC Chapter 7.

Purpose

The maintenance of assemblies required to be fire-resistance rated is a key component in a passive fire protection philosophy. This chapter reinforces this component and further regulates specific protection elements that, when properly maintained, prevent fire spread and smoke migration through buildings.

SECTION 701—GENERAL

701.1 Scope. The provisions of this chapter shall govern the inspection and maintenance of the materials, systems and assemblies used for structural *fire resistance*, *fire-resistance-rated* construction separation of adjacent spaces and construction installed to resist the passage of smoke to safeguard against the spread of fire and smoke within a building and the spread of fire to or from buildings. New buildings shall comply with the *International Building Code.*

C This section establishes the scope of Chapter 7, introduces the requirements for maintaining the integrity of fire-resistance-rated assemblies (see Section 703) and prescribes the types of floor opening protection required in existing buildings (see Section 704). The provisions of Chapter 7 apply to the ongoing maintenance of the materials, assemblies and systems used to protect against the passage of fire and smoke within and between buildings. The assemblies outlined herein provide various degrees of protection. The required fire-resistance rating varies with the potential fire hazard associated with type of construction, occupancy, height and area of the building and degree of protection for different elements of the means of egress. The potential fire hazard associated with various occupancies is reflected in the IBC. IBC Chapter 7 provides the details and the extent of the protection (horizontal and vertical continuity); however, the actual fire-resistance-rated construction is mandated by provisions in IBC Chapters 4, 5, 6, 7 and 10.

701.2 Fire-resistance-rated construction. The *fire-resistance rating* of the following *fire-resistance-rated* construction shall be maintained:

1. Structural members.
2. *Exterior walls*.
3. *Fire walls, fire barriers, fire partitions*.
4. *Horizontal assemblies*.
5. Shaft enclosures.

C Fire walls, fire barriers and fire partitions are key components in a passive fire- and life-safety design. Fire walls constructed in accordance with IBC Section 706 serve to create separate buildings (see the commentary to the definition of "Area, building" in Chapter 2 of the code); therefore, all applicable provisions of the code are applied individually to the building on each side of the fire wall. As such, the fire wall must also provide the same protection afforded by exterior walls, namely: structural integrity, structural independence and adequate fire resistance for exposure protection.

Fire barriers constructed in accordance with IBC Section 707 provide a higher degree of protection than fire partitions, but lack the inherent structural integrity of fire walls and, unlike fire partitions, there are no circumstances under which a fire barrier wall is permitted to terminate at a ceiling. Fire barriers are used to separate a variety of areas, including exits and certain areas of refuge, mixed occupancies and incidental use areas, shafts, floor opening enclosures, hazardous materials control areas and fire areas. It is important to note that since fire barriers are intended to provide a reliable subdivision of areas, the construction that structurally supports the assembly is required to provide and maintain at least the same hourly fire-resistance rating as the fire barrier being supported.

Fire partitions constructed in accordance with IBC Section 708 are wall assemblies that enclose exit access corridors; separate tenant spaces in covered malls, dwelling units and sleeping units; and separate elevator lobbies from the balance of a floor. Openings in fire partitions must be properly protected, but the total area of openings in a fire partition is not limited. Unlike the continuity requirements for fire walls and fire barriers, fire partition continuity must only be continuous from floor slab to the floor slab or roof deck above or to the ceiling of a fire-resistance-rated floor/ceiling or roof/ceiling assembly. Although fire partitions must normally be supported by construction having a comparable fire-resistance rating in buildings of Types IIB, IIIB and VB construction, as defined in the IBC, Section 708.4 of that code does not require such supportive construction for sleeping units and tenant separations and exit access corridor walls.

The IBC includes requirements for fire-resistance rating, continuity and opening and penetration protection in these types of assemblies. Because these assemblies and their opening protectives are critical life safety components of a building, they must be maintained throughout the life of the building. Opening protective maintenance provisions for these types of assemblies are contained in NFPA 80. For further information on fire walls, fire barriers and fire partitions, see the commentaries to IBC Sections 706, 707 and 708, respectively.

Vertical openings that are not properly protected can act as a chimney for smoke, hot gases and products of combustion. Unprotected floor openings have been a major contributing factor in many large loss-of-life fires. Unless indicated otherwise, Chapter 11 retroactively requires the enclosure of vertical openings between floors with approved fire barriers (see commentaries, Sections 1103.4 through 1103.4.10).

701.2.1 Hanging displays. The hanging and displaying of salable goods and other decorative materials from acoustical ceiling systems that are part of a *fire-resistance-rated horizontal assembly* shall be prohibited.

C This section is only applicable to acoustical ceiling systems that are a component of an approved fire-resistance-rated floor/ceiling or roof/ceiling assembly required to be rated by the type of construction of the building. Fire-resistance-rated floor/ceiling and roof/ceiling assemblies must be tested using the methods in ASTM E119 to demonstrate a fire-resistance rating. Locating a substantial fuel load and additional weight directly beneath an acoustical ceiling, however, may expose the ceiling to a direct fire source and weight overload not contemplated in the ASTM E119 testing. That could breech the ceiling, which is an integral part of the tested assembly. Depending on the contribution of the ceiling to the overall fire-resistance rating, this may result in the assembly not functioning as intended or failing completely.

New acoustical ceiling systems, whether or not they are a component of an approved fire-resistance-rated assembly, are required to comply with IBC Section 808. IBC Section 808.1.1 requires that acoustical ceiling systems comply with the manufacturer's installation instructions. IBC Section 808.1.1.1 further requires compliance with several ASTM standards that govern the installation of such systems. Those standards do not contemplate the addition of loads to the metal support framework of the system beyond the load of the system itself plus light fixtures, or other components that might be part of the approved design. Adding any weight to the system beyond that for which it was designed or approved by the building official could lead to failure of the system and should only be done after a review of the structural components by a registered design professional.

701.3 Smoke barriers. The *fire-resistance rating* and smoke-resistant characteristics of *smoke barriers* shall be maintained.

C Smoke barriers divide areas of a building into separate smoke compartments to create an area of safety for occupants. A smoke barrier is designed and installed in accordance with IBC Section 709 to resist fire and smoke spread so that occupants can be evacuated or relocated to adjacent smoke compartments (see the commentaries to the definitions of "Smoke barrier" and "Smoke compartment" in Chapter 2). This concept has proven effective in Group I-2 and I-3 occupancies, and IBC Sections 407.5 and 408.6 identify where smoke barriers are required in those occupancies. Smoke barriers may also be utilized in other applications, such as part of a smoke control system (see Section 909.5 of this code), separation of accessible areas of refuge in accessible means of egress (see Section 1009.6.4 of this code), compartmentation of underground buildings (see IBC Section 405.4.2) and elevator lobbies in underground buildings (see IBC Section 405.4.3), fire service access elevator lobbies (IBC Section 3007.6.2) and occupant evacuation elevator lobbies (IBC Section 3008.6.2) among others.

Other than the wall itself, all of the elements in the smoke barrier that can potentially allow smoke travel through the smoke barrier are required to have a quantified resistance to leakage. This includes doors, joints, through penetrations and dampers. The maximum leakage limits are as established in the individual code sections referenced above for each element. A smoke barrier is not intended or expected to be exposed to fire for extended periods and is, therefore, not required to have a fire-resistance rating exceeding 1 hour. Also, the occupancies in which smoke barriers are required are generally required to be sprinklered by Section 903 of this code. Smoke barriers are to be continuous from outside wall to outside wall and from the top of the foundation or floor/ceiling assembly below to the underside of the floor or roof sheathing, deck or slab. The provisions require the barrier to be continuous through all concealed and interstitial spaces, including suspended ceilings and the space between the ceiling and the floor or roof sheathing, deck or slab above. Smoke barriers are not required to extend through interstitial spaces if the space is designed and constructed such that fire and smoke will not spread from one smoke compartment to another; therefore, the construction assembly forming the bottom of the interstitial space must provide the required fire-resistance rating and be capable of resisting the passage of smoke from the spaces below.

701.4 Smoke partitions. The smoke-resistant characteristics of smoke partitions shall be maintained.

C Smoke partitions are nonrated walls that serve to resist the spread of fire and the unmitigated movement of smoke for an unspecified period of time. Their primary purpose is to prevent the ready and quick passage of smoke into corridors in Group I-2 occupancies or for elevator lobby protection in a sprinklered building. Unlike 1-hour fire-resistance-rated smoke barriers, unless required by the IBC, smoke partitions are not required to have a fire-resistance rating. Smoke partitions are intended to provide less protection than a smoke barrier, and therefore are not required to be continuous through the concealed spaces and through the ceiling. The construction of a smoke partition is prescribed in IBC Section 710; however, the level of performance or a method of testing them is not provided.

Because these assemblies and their opening protectives are critical life safety components of a building, they must be maintained throughout the life of the building. Opening protective maintenance provisions for these types of assemblies are contained in NFPA 105. For further information on incidental uses, smoke barriers and smoke partitions, see the commentaries to IBC Sections 509, 709 and 710, respectively.

701.5 Maintaining protection. Materials, systems and devices used to repair or protect breaches and openings in *fire-resistance-rated* construction and construction installed to resist the passage of smoke shall be maintained in accordance with Sections 703 through 707.

C This section provides a general reference to the sections of this chapter that provide requirements that are used to maintain, repair or protect breaches and openings in fire-resistance-rated construction and the construction installed to resist the passage of smoke.

701.6 Owner's responsibility. The *owner* shall maintain an inventory of all required *fire-resistance-rated* construction, construction installed to resist the passage of smoke and the construction included in Sections 703 through 707 and Sections 602.4.1 and 602.4.2 of the *International Building Code*. Such construction shall be visually inspected by the *owner* annually and properly repaired, restored or replaced where damaged, altered, breached or penetrated. Where concealed, such elements shall not be required to be visually inspected by the *owner* unless the concealed space is accessible by the removal or movement of a panel, access door, ceiling tile or similar movable entry to the space.

C Maintaining an inventory is critical to the owner, the fire code official, and anyone who may be providing the inspection service. This inventory record is necessary for these individuals to perform a complete inspection of the existing protection elements and allow for the continuing documentation that they are still in place and able to perform their required function.

701.6.1 Recordkeeping. Records of all required system inspections, testing, repairs and maintenance shall be maintained in accordance with Section 110.3.

C This section provides a direct reference to Section 110.3 for IFC general recordkeeping requirements. Refer to this section and commentary for further information.

701.7 Unsafe conditions. Where any components in this chapter are not maintained and do not function as intended or do not have the *fire resistance* or the resistance to the passage of smoke required by the code under which the building was constructed, remodeled or altered, such component(s) or portion thereof shall be deemed an unsafe condition, in accordance with Section 115.1.1. Components or portions thereof determined to be unsafe shall be repaired or replaced to conform to that code under which the building was constructed, remodeled or altered or this chapter, as deemed appropriate by the *fire code official*.

Where the condition of components is such that any building, structure or portion thereof presents an imminent danger to the occupants of the building, structure or portion thereof, the *fire code official* shall act in accordance with Section 115.2.

C This section is intended to clarify to code officials, designers, contractors and property owners that a building's fire-resistance-rated construction must be maintained at the original level of safety required by the codes that were applicable when the building was constructed or last remodeled. Failure to maintain fire-resistant components to that level of safety will result in the component being declared unsafe in accordance with Section 115.1.1 and repair or restoration being required.

Code provisions, which require maintenance to a level of safety required by an often-unknown code or an unknown edition of a known code, are sometimes viewed as problematic and unenforceable. However, communities should have some record of when a building was constructed, and knowing the year of construction should make it relatively easy to determine an edition of the code that was published close to or prior to that year. In many communities, local historical societies can also be helpful in doing research to determine the year of construction. These types of methods for determining the originally applicable code, if any, could be viewed as haphazard and arbitrary, but they can be considered better than trying to make a building constructed 30, 50 or 100 years ago comply with today's requirements. In the event that no information of any kind can be found to shed light on an original construction date, this chapter provides for an acceptable level of safety and can be retroactively required where deemed appropriate by the fire code official.

This section also provides that where component conditions are so bad due to lack of maintenance as to constitute a clear and present threat to the safety of the occupants, the fire code official must take the steps required by Section 115.2.

SECTION 702—DEFINITIONS

702.1 Definitions. The following terms are defined in Chapter 2:

DRAFTSTOP.

FIREBLOCKING.

MEMBRANE-PENETRATION FIRESTOP SYSTEM.

OPENING PROTECTIVE.

SMOKE BARRIER.

SMOKE PARTITION.

THROUGH-PENETRATION FIRESTOP SYSTEM.

C Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2 or in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (see also commentaries, Sections 201.1 through 201.4).

SECTION 703—PENETRATIONS

703.1 Maintaining protection. Materials and firestop systems used to protect membrane and through penetrations in *fire-resistance-rated* construction and construction installed to resist the passage of smoke shall be maintained. The materials and firestop systems shall be securely attached to or bonded to the construction being penetrated with no openings visible through or into the cavity of the construction. Where the system design number is known, the system shall be inspected to the listing criteria and manufacturer's installation instructions.

C The code mandates that all equipment, systems, devices and safeguards required by the current and previously adopted codes be maintained in good working order (see Section 102.1). This section reiterates that requirement specifically for fire-resistance-rated assemblies in existing buildings.

Once a building is occupied, its component parts are often damaged, altered or penetrated for installation of new piping, wiring and the like. These actions may reduce the effectiveness of assemblies that must be fire-resistance rated. This section requires the building owner, annually, to visually inspect nonconcealed elements and that any damage to a fire-resistance-rated assembly be repaired in a manner that restores the original required performance characteristics. Concealed elements must be visually inspected if they may be accessed by a door, removable ceiling tile, access panel or the like. Similarly, if a fire-resistance-rated assembly is altered or penetrated, the alteration or penetration must comply with the applicable requirements of the IBC for the particular type of alteration or penetration.

This section also requires that written records of maintenance and repairs to rated assemblies must be kept and should indicate the date, time and the name of the person conducting the inspection or repair for each rated assembly. These records must be maintained by the owner and made available to the fire code official for review when requested. This requirement relieves the fire code official of the administrative burden of maintaining test records. Note that, if the owner does not have the design number, the base inspection criteria would still apply, inspecting to make sure the system is properly secured and inspecting for visible openings through the system or into the cavity.

703.2 Repair of penetrations. Where damaged, materials used to protect membrane- and through-penetrations shall be replaced or restored with materials or systems that meet or exceed the code requirements applicable at the time when the assembly was constructed, remodeled or altered.

C This section provides specific language to clarify the intent of Section 703.1. Where protection of penetrations and joints has been damaged, it needs to be repaired, not simply "maintained" in a damaged condition. The code recognizes that penetrations and joints that were protected in accordance with previous code editions may not have used the types of tested systems that are mandated in current building codes. Consequently, the section continues the allowance to grandfather installed penetration and joint seals when they are damaged but requires that they be repaired to meet the requirements that were applicable when the penetration or joint protection was installed.

SECTION 704—JOINTS AND VOIDS

704.1 Maintaining protection. Where required when the building was originally constructed, materials and systems used to protect joints and voids in the following locations shall be maintained. The materials and systems shall be securely attached to or bonded to the adjacent construction, without openings visible through the construction.

1. Joints in or between *fire-resistance-rated* walls, floors or floor/ceiling assemblies and roof or roof/ceiling assemblies.
2. Joints in *smoke barriers*.
3. Voids at the intersection of a horizontal floor assembly and an exterior curtain wall.
4. Voids at the intersection of a horizontal *smoke barrier* and an exterior curtain wall.
5. Voids at the intersection of a nonfire-resistance-rated floor assembly and an exterior curtain wall.

6. Voids at the intersection of a vertical *fire barrier* and an exterior curtain wall.
7. Voids at the intersection of a vertical *fire barrier* and a nonfire-resistance-rated roof assembly.

Unprotected joints and voids do not need to be protected where such joints and voids were not required to be protected when the building was originally constructed. Where the system design number is known, the system shall be inspected to the listing criteria and manufacturer's installation instructions.

C This section lists the specific locations where fire and smoke protection joints and voids need to be maintained. In general, these occur where two individual building protection elements come together and those elements need to be continually connected at their intersecting junction. There should be no visible openings or any way for fire and smoke to pass through. Note that this section is not intended to be retroactive in the addition of protection elements. As stated, joints and voids that were not required to be protected when the building was originally built are not required to have protection added as part of the building maintenance. However, if the listing information is available, the maintenance requirements provided by the manufacturer are to be followed.

704.2 Repair of joints and voids. Where damaged, materials used to protect joints and voids shall be replaced or restored with materials or systems that meet or exceed the code requirements applicable at the time when the assembly was constructed, remodeled or altered.

C Similar to Section 703.2, this section provides specific language to clarify the intent of Section 704.1. Where protection of joints and voids is damaged, it needs to be repaired, not simply "maintained" in a damaged condition. The code recognizes that joints and voids that were protected in accordance with previous code editions may not have used the types of tested systems that are mandated in current building codes. Consequently, the section continues the allowance to grandfather installed systems when they are damaged but requires that they be repaired to meet the requirements that were applicable when the protection was installed.

704.3 Opening protectives. Where openings are required to be protected, opening protectives shall be maintained self-closing or automatic-closing by smoke detection. Existing fusible-link-type automatic door-closing devices are permitted if the fusible link rating does not exceed 135°F (57°C).

C This section requires that fire door assemblies provided for protection of openings in vertical enclosures be self-closing or automatic-closing in order to maintain the integrity of the vertical opening enclosure. This section also recognizes that some opening protectives in existing buildings may already be equipped with heat-actuated closing devices rather than the smoke-detector-actuated devices otherwise required by the section. Such devices are allowed to continue in service, provided that the temperature rating of their fusible element is as low as is available [i.e., 135°F (57°C)] to provide the fastest possible operation in the event of a fire. In the event that an existing fusible link on an opening protective is rated higher than the maximum 135°F (57°C) allowed by this section, it would need to be removed and the door maintained as self-closing or be replaced with a smoke-detector-actuated closer in accordance with this section and Section 907.3. New opening protectives must comply with IBC Section 716. See the commentary to that section for further information.

SECTION 705—DOOR AND WINDOW OPENINGS

705.1 General. Where required when the building was originally constructed, opening protectives installed in *fire-resistance-rated* assemblies, *smoke barriers* and *smoke partitions* shall be inspected and maintained in accordance with this section.

C This section provides a general reference to ensuing sections that list specific requirements for the maintenance of existing opening protectives that were required and installed as part of fire and smoke protection construction when the building was originally built.

705.2 Inspection and maintenance. *Opening protectives* in *fire-resistance-rated* assemblies shall be inspected and maintained in accordance with NFPA 80. *Opening protectives* in *smoke barriers* shall be inspected and maintained in accordance with NFPA 80 and NFPA 105. Openings in smoke partitions shall be inspected and maintained in accordance with NFPA 105. Fire doors and smoke and draft control doors shall not be blocked, obstructed, or otherwise made inoperable. Fusible links shall be replaced promptly whenever fused or damaged. *Opening protectives* and smoke and draft control doors shall not be modified.

C Openings in fire-resistance-rated assemblies must be protected to prevent the passage of fire in accordance with IBC Section 716. After opening protectives are installed and approved, they may become damaged, corroded or otherwise less effective than required. This section specifically requires that all opening protectives required by the IBC be maintained in compliance with NFPA 80 so that they can perform their intended function, which is to prevent the passage of smoke, fire or combustion products through openings in fire-resistance-rated walls, ceilings and shafts during a fire emergency. Sections 705.2.3 and 705.2.4 of the code indicate specific points of inspection and enforcement regarding these doors. Prohibited modifications to fire door assemblies include the attachment of materials, cutting, boring holes or other alterations that could affect the performance of the door as a fire protection-rated assembly.

Proper maintenance necessitates that the manufacturer's installation instructions and the listing organization's instructions are followed in order to maintain the listing and proper operation of the assemblies and devices as required by the code, the manufacturer and the listing organization.

705.2.1 Labeling requirements. Where *approved* by the *fire code official*, the application of field-applied labels associated with the maintenance of *opening protectives* shall follow the requirements of the *approved* third-party certification organization accredited for listing the *opening protective*.

C This section addresses the very real issue of maintaining labeled opening protectives by requiring field-applied labels to follow the requirements of the third-party certification organization, which is accredited for listing the specific opening protective. The relabeling of existing fire doors is a common practice and, due to the importance of the rating requirements, a level of monitoring by a third party to ensure the labeling matches the rating of the door assembly is necessary.

In the listing documentation, there are specific criteria for field application of labels. One of the criteria is whether the local fire code official allows this practice, and this section provides guidance in this area to the fire code official. IBC Section 716.2.9.1 requires that new fire doors or new fire door assemblies must be labeled at the factory (see the definition of "Labeled" in Chapter 2 of this code). However, it is not uncommon for an existing fire door to have either a damaged or missing label, or a label that has been painted over or otherwise obscured. The fire code official needs to make a determination as to whether field application of the label is acceptable or not. If field application is allowed, then the certification organization can follow the proper criteria for labeling the opening protective.

705.2.2 Signs. Where required by the *fire code official*, a sign shall be permanently displayed on or near each *fire door* in letters not less than 1 inch (25 mm) high to read as follows:

1. For doors designed to be kept normally open: "FIRE DOOR—DO NOT BLOCK."
2. For doors designed to be kept normally closed: "FIRE DOOR—KEEP CLOSED."

C Any door in a fire-resistance-rated wall assembly represents a potential "weak link" in maintaining the degree of compartmentation intended. That is the reason for requiring a rated assembly. The IBC calls for adequate opening protection in the form of a door with a specified fire protection rating. This section allows the fire code official to require signage on or near the rated doors to make the occupants aware of the importance of the door as a fire- and life-safety feature. Also, see the commentary to Section 705.2.3 for a discussion on door closing and the improper use of props to hold doors open.

705.2.3 Hold-open devices and closers. Hold-open devices and automatic door closers, where provided, shall be maintained. During the period that such device is out of service for repairs, the door it operates shall remain in the closed position.

C The only devices acceptable for holding fire doors open are fire-detector-activated, automatic-closing devices that automatically close the doors (or allow the doors to swing closed using self-closing devices) in the event of a fire. Numerous devices, such as electromagnetic hold-opens, pneumatic systems and systems of pulleys and weights connected to a fusible link, are available.

The detection method for the closing device must be consistent with the purpose of the opening protective; that is, doors in smoke barriers must be activated by smoke detectors. Heat detectors or fusible links are adequate where maintenance of the fire-resistance rating alone is required.

Where smoke-detector-activated automatic door closers are used and the detectors are interconnected with a required fire alarm system, the devices and wiring methods must be checked for compatibility with the fire alarm system control panel before installation. Some fire alarm control equipment is compatible only with the manufacturer's automatic smoke detectors. Fire detectors used for automatic door release service in buildings that are not equipped with a fire alarm system must comply with Section 907.3.

A common violation of fundamental safety principles, as well as this section of the code, is having wooden or rubber wedges, or kick-down-type door hold-opens prop open fire doors or smoke barrier doors. This renders them totally ineffective as opening protectives. Building maintenance personnel who do not understand the purpose of these doors often do this to aid movement of people, equipment or air in a hallway or other area without realizing the potential hazard to life safety if a fire were to occur. This violation is especially problematic as it pertains to means of egress stairwells or horizontal exit doors as well. For further information on door closing requirements for fire doors, see IBC Section 716.2.6.

705.2.4 Door operation. Swinging *fire doors* shall close from the full-open position and latch automatically.

C Fire doors must be closed to be effective. Swinging fire doors should be frequently checked to make sure they close and latch on their own power from any position.

705.2.5 Smoke- and heat-activated doors. Smoke-activated doors shall be maintained to self-close or automatically close upon detection of smoke. Existing fusible-link-type automatic door-closing devices are permitted if the fusible link rating does not exceed 135°F (57°C).

C This section requires that fire door assemblies provided for protection of openings in vertical enclosures be self-closing or automatic-closing in order to maintain the integrity of the vertical opening enclosure. This section also recognizes that some opening protectives in existing buildings may already be equipped with heat-actuated closing devices rather than the smoke-detector-actuated devices otherwise required by the section. Such devices are allowed to continue in service, provided that the temperature rating of their fusible element is as low as is available [i.e., 135°F (57°C)] to provide the fastest possible operation in the event of a fire. In the event that an existing fusible link on an opening protective is rated higher than the maximum 135°F (57°C) allowed by this section, it would need to be removed and the door maintained as self-closing or be replaced with a smoke-detector-actuated closer in accordance with this section and Section 907.3. New opening protectives must comply with IBC Section 716. See the commentary to that section for further information.

705.2.6 Testing. Horizontal and vertical sliding and rolling *fire doors* shall be inspected and tested annually to confirm proper operation and full closure. Records of inspections and testing shall be maintained.

C Annual tests are intended to determine that required fire doors operate freely and close completely. Where fusible links are used as the releasing mechanism, the link may be temporarily removed rather than activated during testing. Fusible links in poor condition must be replaced as part of the maintenance of fire-resistance components. Smoke detectors and heat detectors other than fusible links must be tested as required by the manufacturer's instructions (see NFPA 72 for recommended testing procedures for various fire detectors).

This section also requires that written records of inspection and testing of opening protectives must be kept and should indicate the date, the time and the name of the person conducting the inspection or repair for each rated assembly. These records must be maintained by the owner and should be made available to the fire code official for review when requested. This requirement relieves the fire code official of the administrative burden of maintaining test records.

705.2.7 Periodic inspection and testing of rolling steel fire doors. Rolling steel *fire doors* shall be inspected and tested annually by a trained rolling steel *fire door* systems technician in accordance with the applicable provisions of NFPA 80. Records of inspections and testing shall be maintained.

C NFPA 80 includes requirements for the periodic inspection and testing of rolling fire doors and provides additional clarity on the qualifications of inspection and testing personnel. Specific product training for rolling fire doors should include a visual inspection of the door assembly, a check of the door operation, a drop test and resetting of the door to operational status. The complex nature of these products dictates that personnel be familiar with the type and feature of the door assembly, as well as manufacturer-specific details necessary to help ensure the door will function as intended and as required by the code.

NFPA 80 uses the defined term "Trained rolling steel fire door systems technician" to indicate that personnel conducting the inspections be specifically trained for these systems. This is consistent with the recommendations in Door and Access Systems Manufacturers Association (DASMA) Technical Data Sheet #271, which states that door drop testing be conducted by a "trained door systems technician."

Section 705 contains no other provisions for the training of inspection personnel; this section references the NFPA 80 provisions to help ensure that inspections, testing and resetting of rolling fire doors are conducted by qualified technicians and that the rolling fire doors in use as part of fire-resistance-rated assemblies will perform as intended.

SECTION 706—DUCT AND AIR TRANSFER OPENINGS

706.1 Maintaining protection. Dampers protecting ducts and air transfer openings shall be inspected and maintained in accordance with NFPA 80 and NFPA 105. Other products or materials used to protect the openings for ducts and air transfer openings shall be securely attached to or bonded to the construction containing the duct or air transfer opening, without visible openings through or into the cavity of the construction. Any damaged products or materials protecting duct and air transfer openings shall be repaired, restored or replaced.

C This section addresses the maintenance of dampers in ducts and air transfer openings. The referenced NFPA standards provide the details of the required inspections. NFPA 80 is specific to fire doors and NFPA 105 is specific to smoke doors. Furthermore, any other products or materials that are installed to protect openings in ducts or air transfer openings are to be inspected to ensure that they continue to be firmly attached. There should be no visible openings or any way for fire and smoke to pass through. If the inspection discovers any components that are damaged or otherwise not performing their intended function, they are to be repaired or replaced as necessary.

706.2 Unprotected openings. Unprotected duct and air transfer openings in *fire-resistance-rated* construction and construction installed to resist the passage of smoke shall be protected so as to comply with requirements that were in effect when the building was constructed.

C This section makes it clear that unprotected duct and air transfer openings in fire and smoke protected construction are only required to be protected with the requirements that were in effect when the building was originally built.

SECTION 707—CONCEALED SPACES

707.1 Fireblocking and draftstopping. Required *fireblocking* and *draftstopping* in combustible concealed spaces shall be maintained to provide continuity and integrity of the construction.

C Fireblocking and draftstopping (see the Chapter 2 commentaries for the definitions "Fireblocking" and "Draftstop") slow the spread of fire and the products of combustion through concealed spaces within a building. To fulfill their intended function, fireblocking and draftstopping must be properly maintained. Most frequently, damage or repairs to other building components, such as mechanical piping, results in fireblocking or draftstopping being removed and not properly replaced. This section specifically requires that where fireblocking and draftstopping required by the IBC are damaged, removed or otherwise altered, they must be replaced or restored.

SECTION 708—SPRAY FIRE-RESISTANT MATERIALS AND INTUMESCENT FIRE-RESISTANT MATERIALS

708.1 Maintaining protection. Where required when the building was originally permitted and constructed, spray fire-resistant materials and intumescent fire-resistant materials shall be visually inspected to verify that the materials do not exhibit exposure to the substrate.

C This section provides the specific maintenance requirement for these types of materials if they were required to be installed when the building was originally permitted. As noted, the requirement is that the materials are to be visually inspected as part of the maintenance procedure to determine if there is any exposed substrate, which would indicate that the protection in no longer fully in place.